General Relativity
Final Examination — Answers

Conventions. Signature $(-,+,+,+)$, units $c = 1$. Covariant derivatives are $\nabla_\mu V^\rho = \partial_\mu V^\rho + \Gamma_{\mu\nu}{}^\rho V^\nu$ and $\nabla_\mu \omega_\nu = \partial_\mu \omega_\nu - \Gamma_{\mu\nu}{}^\rho \omega_\rho$, i.e. the first lower index of $\Gamma_{\mu\nu}{}^\rho$ is the derivative index. Primes denote quantities in new coordinates x'^μ.

Question 1. *(2 points)*

Under a change of coordinates $x^\mu \to x'^\mu(x)$ the Riemann tensor, being a $(1,3)$ tensor, transforms as

$$R'^\mu{}_{\nu\rho\sigma} = \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x^\delta}{\partial x'^\sigma} R^\alpha{}_{\beta\gamma\delta}\,. \tag{1}$$

The Ricci tensor in the new coordinates is $R'_{\nu\sigma} = R'^\rho{}_{\nu\rho\sigma}$, i.e. set $\mu = \rho$ in (1) and sum:

$$\begin{aligned} R'_{\nu\sigma} = R'^\rho{}_{\nu\rho\sigma} &= \frac{\partial x'^\rho}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x^\delta}{\partial x'^\sigma} R^\alpha{}_{\beta\gamma\delta} \\ &= \left(\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x'^\rho}{\partial x^\alpha}\right)\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma} R^\alpha{}_{\beta\gamma\delta}\,. \end{aligned} \tag{2}$$

By the chain rule,

$$\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x'^\rho}{\partial x^\alpha} = \frac{\partial x^\gamma}{\partial x^\alpha} = \delta^\gamma_\alpha\,, \tag{3}$$

(the Jacobian matrices of the transformation and its inverse multiply to the identity). Hence

$$R'_{\nu\sigma} = \frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma} R^\alpha{}_{\beta\alpha\delta} = \frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma} R_{\beta\delta}\,, \tag{4}$$

which is exactly the transformation law of a $(0,2)$ tensor. So the Ricci tensor is a tensor: contraction of an upper index with a lower index of a tensor gives a tensor because the factors of the Jacobian and inverse Jacobian cancel.

Question 2. *(3 points)*

First note that $\hat\Gamma$ is a genuine connection: the difference of two connections is a tensor, and conversely adding the tensor $\frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma} \equiv \frac{1}{12} H_{\mu\nu}{}^\rho$ to the Levi-Civita connection gives an object which transforms with the same inhomogeneous term as Γ, hence a connection. We write $\hat\Gamma_{\mu\nu}{}^\rho = \Gamma_{\mu\nu}{}^\rho + \frac{1}{12} H_{\mu\nu}{}^\rho$.

a) **Torsion.** The torsion tensor is

$$T_{\mu\nu}{}^\rho = \hat\Gamma_{\mu\nu}{}^\rho - \hat\Gamma_{\nu\mu}{}^\rho = \underbrace{\Gamma_{\mu\nu}{}^\rho - \Gamma_{\nu\mu}{}^\rho}_{=0 \text{ (Levi-Civita is symmetric)}} + \frac{1}{12} g^{\rho\sigma}\left(H_{\mu\nu\sigma} - H_{\nu\mu\sigma}\right) = \frac{1}{6} g^{\rho\sigma} H_{\mu\nu\sigma}\,, \tag{5}$$

using $H_{\nu\mu\sigma} = -H_{\mu\nu\sigma}$. Since $g^{\rho\sigma}$ is invertible this vanishes only if $H_{\mu\nu\rho} = 0$. So for $H \neq 0$ the connection $\hat{\Gamma}$ is **not torsion-free**; its torsion is $T_{\mu\nu\rho} = \frac{1}{6}H_{\mu\nu\rho}$.

Metric compatibility. Using the rule for the covariant derivative of a $(0,2)$ tensor,

$$\begin{aligned}\hat{\nabla}_\mu g_{\nu\lambda} &= \partial_\mu g_{\nu\lambda} - \hat{\Gamma}_{\mu\nu}{}^{\rho} g_{\rho\lambda} - \hat{\Gamma}_{\mu\lambda}{}^{\rho} g_{\nu\rho} \\ &= \underbrace{\partial_\mu g_{\nu\lambda} - \Gamma_{\mu\nu}{}^{\rho} g_{\rho\lambda} - \Gamma_{\mu\lambda}{}^{\rho} g_{\nu\rho}}_{=\nabla_\mu g_{\nu\lambda}=0} - \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma} g_{\rho\lambda} - \frac{1}{12} g^{\rho\sigma} H_{\mu\lambda\sigma} g_{\nu\rho} \\ &= -\frac{1}{12}\left(H_{\mu\nu\lambda} + H_{\mu\lambda\nu}\right) = 0\,, \end{aligned} \tag{6}$$

where we used $g^{\rho\sigma} g_{\rho\lambda} = \delta^\sigma_\lambda$, that the Levi-Civita connection is metric compatible, and the antisymmetry $H_{\mu\lambda\nu} = -H_{\mu\nu\lambda}$. Hence $\hat{\Gamma}$ **is metric compatible**.

(If one instead uses the convention in which the *second* lower index is the derivative index, the extra terms become $-\frac{1}{12}(H_{\nu\mu\lambda} + H_{\lambda\mu\nu}) = -\frac{1}{12}(H_{\nu\mu\lambda} - H_{\nu\mu\lambda}) = 0$, using total antisymmetry, and the torsion changes only by an overall sign; the conclusions are unchanged.)

So $\hat{\Gamma}$ is a metric-compatible connection with totally antisymmetric torsion; it differs from the unique torsion-free metric-compatible (Levi-Civita) connection precisely by having torsion.

b) The geodesic equation for a curve $x^\rho(\lambda)$ with respect to $\hat{\Gamma}$ is the statement that the tangent vector is parallel transported along itself, $\frac{dx^\mu}{d\lambda}\hat{\nabla}_\mu \frac{dx^\rho}{d\lambda} = 0$, i.e.

$$\frac{d^2x^\rho}{d\lambda^2} + \hat{\Gamma}_{\mu\nu}{}^{\rho}\,\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0\,. \tag{7}$$

Inserting $\hat{\Gamma}_{\mu\nu}{}^{\rho} = \Gamma_{\mu\nu}{}^{\rho} + \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma}$:

$$\frac{d^2x^\rho}{d\lambda^2} + \Gamma_{\mu\nu}{}^{\rho}\,\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} + \frac{1}{12} g^{\rho\sigma} H_{\mu\nu\sigma}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0\,. \tag{8}$$

The product $\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda}$ is symmetric in $\mu \leftrightarrow \nu$ while $H_{\mu\nu\sigma}$ is antisymmetric in $\mu \leftrightarrow \nu$, so

$$H_{\mu\nu\sigma}\dot{x}^\mu\dot{x}^\nu = H_{\nu\mu\sigma}\dot{x}^\nu\dot{x}^\mu = -H_{\mu\nu\sigma}\dot{x}^\mu\dot{x}^\nu \quad \Rightarrow \quad H_{\mu\nu\sigma}\dot{x}^\mu\dot{x}^\nu = 0\,, \tag{9}$$

(relabelling dummy indices in the first step). Therefore the geodesic equation reduces to

$$\frac{d^2x^\rho}{d\lambda^2} + \Gamma_{\mu\nu}{}^{\rho}\,\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0\,, \tag{10}$$

which is the geodesic equation of the Levi-Civita connection. The two connections therefore have identical geodesics (with the same affine parameters). More generally, only the symmetric part of a connection enters the geodesic equation, and $\hat{\Gamma}_{(\mu\nu)}{}^{\rho} = \Gamma_{\mu\nu}{}^{\rho}$.

Question 3. *(4 points)*

a) The metric components blow up or degenerate at:

- $r = 2GM$ (the Schwarzschild radius / event horizon), where $g_{tt} = -f \to 0$ and $g_{rr} = f^{-1} \to \infty$;
- $r = 0$, where $g_{tt} = -f \to +\infty$ and $g_{rr} \to 0$;
- (also $\theta = 0, \pi$, where $g_{\phi\phi} = r^2 \sin^2\theta = 0$; this is just the familiar failure of spherical polar coordinates at the poles of the sphere and is harmless.)

To decide whether a singularity is physical we must look at coordinate-invariant quantities, e.g. curvature scalars. For Schwarzschild $R_{\mu\nu} = 0$, so $R = 0$, but the Kretschmann scalar is

$$K = R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} = \frac{48 G^2 M^2}{r^6} \,. \tag{11}$$

At $r = 2GM$ this is finite, $K = 3/(4G^4M^4)$, so nothing physically singular happens there: it is a **coordinate singularity**, caused by the bad behaviour of the (t, r) coordinates (the Schwarzschild time $t \to \infty$ along an infalling worldline as it approaches the horizon). An infalling observer crosses $r = 2GM$ in finite proper time and, by the equivalence principle, notices nothing locally special there (although it is the point of no return: inside, r is a timelike coordinate and decreases inexorably). For a large (e.g. supermassive) black hole, the tidal accelerations at the horizon, $\sim 2GM\ell/r^3 = \ell/(4G^2M^2)$ for a body of size ℓ, are small.

At $r = 0$, $K \to \infty$: this is a genuine **curvature singularity**. Tidal forces (governed by the Riemann tensor through the geodesic deviation equation) diverge there, so any extended body is stretched radially and compressed transversely without bound ("spaghettification"), and spacetime ceases to be described by GR. This is the singularity you should genuinely be concerned about — and once you are inside the horizon you cannot avoid it: you will reach $r = 0$ within a proper time of at most πGM after crossing $r = 2GM$, since $r = 0$ lies in your future, not at a place in space.

b) Differentiate the definition of v:

$$\begin{aligned} dv &= dt + dr + 2GM\, \frac{1/(2GM)}{\frac{r}{2GM} - 1}\, dr = dt + \left(1 + \frac{2GM}{r - 2GM}\right) dr \\ &= dt + \frac{r}{r - 2GM}\, dr = dt + \frac{dr}{f(r)} \,. \end{aligned} \tag{12}$$

(This holds for both $r > 2GM$ and $r < 2GM$ thanks to the absolute value.) Hence $dt = dv - f^{-1} dr$ and

$$\begin{aligned} -f\, dt^2 + f^{-1} dr^2 &= -f \left(dv^2 - \frac{2}{f}\, dv\, dr + \frac{1}{f^2} dr^2 \right) + \frac{1}{f} dr^2 \\ &= -f\, dv^2 + 2\, dv\, dr - \frac{1}{f} dr^2 + \frac{1}{f} dr^2 = -f\, dv^2 + 2\, dv\, dr \,. \end{aligned} \tag{13}$$

The Schwarzschild metric in ingoing Eddington–Finkelstein coordinates is therefore

$$ds^2 = -\left(1 - \frac{2GM}{r}\right) dv^2 + 2\, dv\, dr + r^2 \left(d\theta^2 + \sin^2\theta\, d\phi^2\right) . \tag{14}$$

How this supports a): in these coordinates the metric components are

$$g_{vv} = -\left(1 - \frac{2GM}{r}\right), \quad g_{vr} = g_{rv} = 1, \quad g_{rr} = 0, \quad g_{\theta\theta} = r^2, \quad g_{\phi\phi} = r^2 \sin^2\theta, \tag{15}$$

which are all finite and smooth at $r = 2GM$, and $\det g = -r^4 \sin^2\theta \neq 0$ there, so the metric is non-degenerate and invertible across the horizon. The metric extends smoothly from $r > 2GM$ to $0 < r < 2GM$, confirming that the singularity at $r = 2GM$ was an artefact of the Schwarzschild coordinates. In contrast, $g_{vv} \to \infty$ as $r \to 0$ in *these* coordinates too, consistent with $r = 0$ being a true (coordinate-independent) singularity, as shown by K.

We can also see the causal structure: radial null curves ($d\theta = d\phi = 0$, $ds^2 = 0$) satisfy $dv\ (-f\, dv + 2\, dr) = 0$, giving

$$v = \text{const} \quad \text{(ingoing)}, \qquad \frac{dr}{dv} = \frac{1}{2}\left(1 - \frac{2GM}{r}\right) \quad \text{(“outgoing”)}. \tag{16}$$

Ingoing light rays pass smoothly through $r = 2GM$ at finite v, so infalling observers (who move inside the light cones) do too. For $r < 2GM$ even the “outgoing” rays have $dr/dv < 0$: all future-directed causal curves move to smaller r and end at $r = 0$. Thus $r = 2GM$ is a regular null surface (event horizon) through which you can fall without incident, but you will then inevitably hit the true singularity at $r = 0$.

Question 4. *(5 points)*

a) The metric has the Friedmann–Lemaître–Robertson–Walker (FLRW) form

$$ds^2 = -dt^2 + a(t)^2\, d\Sigma_3^2, \qquad d\Sigma_3^2 = d\chi^2 + \sin^2\chi\, (d\theta^2 + \sin^2\theta\, d\phi^2) , \tag{17}$$

where $d\Sigma_3^2$ is the metric on a unit round three-sphere, i.e. a maximally symmetric space of positive curvature ($k = +1$). Each constant-t slice is therefore homogeneous and isotropic, which is exactly the cosmological principle that our universe appears to satisfy on large scales (e.g. isotropy of the CMB). t is the proper time of comoving observers at fixed (χ, θ, ϕ). The **scale factor** is

$$a(t) = L \cosh(t/L) . \tag{18}$$

It is moreover a solution of Einstein’s equations: with $\dot{a} = \sinh(t/L)$,

$$\frac{\dot{a}^2}{a^2} + \frac{k}{a^2} = \frac{\sinh^2(t/L) + 1}{L^2 \cosh^2(t/L)} = \frac{1}{L^2} , \tag{19}$$

which is the Friedmann equation $H^2 + k/a^2 = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}$ with $\rho = 0$ and a positive cosmological constant $\Lambda = 3/L^2$. Also $\ddot{a} = a/L^2 > 0$, so the expansion is accelerating; for $t \gg L$, $a(t) \approx \frac{L}{2}e^{t/L}$ grows exponentially with Hubble rate $H \to 1/L$ and the curvature term k/a^2 becomes negligible. Observations indicate our universe is now dominated by dark energy behaving like a cosmological constant and is undergoing accelerated expansion, and is spatially very close to flat; it will asymptotically approach de Sitter. A de Sitter phase is also the standard model of inflation in the very early universe. So this is a good candidate cosmological solution (at least for the late-time and inflationary epochs).

b) Let V^μ be any vector at a point. Then

$$g_{\mu\nu}V^\mu V^\nu = \Omega^2\, \bar{g}_{\mu\nu}V^\mu V^\nu\,, \tag{20}$$

and since $\Omega^2 > 0$ (for a non-vanishing, smooth conformal factor), $g(V,V)$ and $\bar{g}(V,V)$ always have the same sign. So a vector is timelike, null or spacelike with respect to g if and only if it is so with respect to $\bar{g}$: the two metrics have identical light cones at every point. Causal structure — which curves are timelike/causal, and hence which points can send signals to which (the causal past and future of every point) — is determined entirely by the light cones, so it is the same for both metrics. (Null geodesics are also the same curves, up to reparametrisation.) Only distances and times differ. This is why we may conformally rescale a spacetime to bring infinity to a finite coordinate distance when drawing a Penrose diagram.

c) With $\cosh(t/L) = 1/\cos\eta$: as t runs over $(-\infty,\infty)$, $\cosh(t/L)$ runs over $[1,\infty)$ and $\cos\eta \in (0,1]$, so $\eta \in (-\pi/2, \pi/2)$ (choosing η to have the same sign as t). Then

$$\sinh(t/L) = \pm\sqrt{\cosh^2(t/L) - 1} = \pm\sqrt{\frac{1}{\cos^2\eta} - 1} = \tan\eta\,. \tag{21}$$

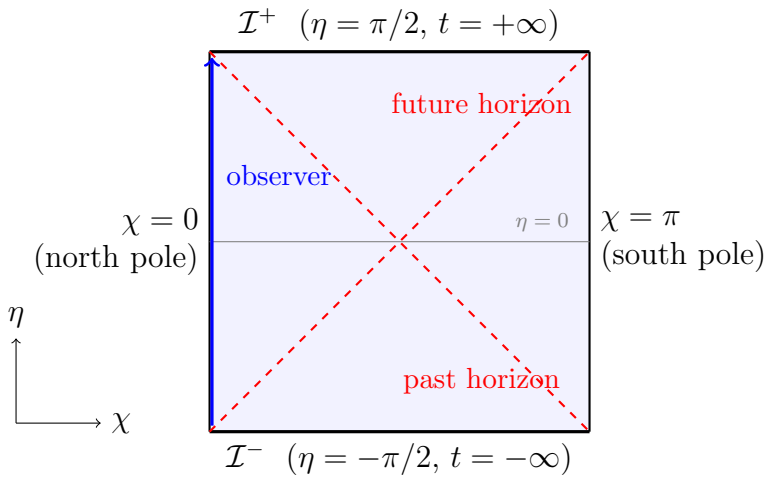
Differentiating the defining relation,

$$\frac{1}{L}\sinh(t/L)\,dt = \frac{\sin\eta}{\cos^2\eta}\,d\eta \quad \Rightarrow \quad \frac{1}{L}\tan\eta\,dt = \frac{\tan\eta}{\cos\eta}d\eta \quad \Rightarrow \quad dt = \frac{L\,d\eta}{\cos\eta}\,. \tag{22}$$

Hence $dt^2 = L^2 d\eta^2/\cos^2\eta$ and $L^2\cosh^2(t/L) = L^2/\cos^2\eta$, so

$$ds^2 = \frac{L^2}{\cos^2\eta}\left(-d\eta^2 + d\chi^2 + \sin^2\chi\left(d\theta^2 + \sin^2\theta\,d\phi^2\right)\right)\,, \qquad \eta \in \left(-\tfrac{\pi}{2},\tfrac{\pi}{2}\right)\,,\ \chi \in [0,\pi]\,. \tag{23}$$

This is conformal, with $\Omega^2 = L^2/\cos^2\eta$, to the Einstein static universe $d\bar{s}^2 = -d\eta^2 + d\chi^2 + \sin^2\chi\, d\Omega_2^2$ (on $\mathbb{R}\times S^3$), restricted to the strip $|\eta| < \pi/2$. By part b) the causal structure is that of this finite region, which we can draw in the (χ,η) plane. Radial null rays satisfy $d\eta = \pm d\chi$, i.e. they are lines at 45°. The conformal factor diverges at $\eta = \pm\pi/2$, which are the (conformal) boundaries at infinite proper time, future and past infinity $\mathcal{I}^\pm$. The Penrose diagram is a **square** of width π (in χ) and height π (in η):



Features of the diagram:

- Each interior point represents a two-sphere of radius $\propto \sin\chi$; the vertical edges $\chi = 0$ and $\chi = \pi$ are the north and south poles of the S^3, where this two-sphere shrinks to a point (they are not boundaries of spacetime).
- $\mathcal{I}^+$ and $\mathcal{I}^-$ (top and bottom) are *spacelike*, in contrast to Minkowski space where null infinity is null.
- Because the height equals the width, a light ray leaving the north pole at $\eta = -\pi/2$ only just reaches the south pole at $\eta = +\pi/2$. Consequently an observer at $\chi = 0$ can never see the whole spacetime: the diagonal from $(\chi, \eta) = (0, \pi/2)$ to $(\pi, -\pi/2)$ is the observer's future event horizon (the boundary of their causal past), and the other diagonal is the past (particle) horizon. Only the left-hand quarter of the square is causally connected to the observer in both directions (the "static patch").

Question 5. *(6 points)*

a) Under an infinitesimal diffeomorphism $x^\mu \to x^\mu - \xi^\mu$, $h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, so $h \to h + 2\partial\cdot\xi$ and

$$\bar{h}_{\mu\nu} \to \bar{h}_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu - \eta_{\mu\nu}\partial^\rho\xi_\rho\,, \qquad \partial^\mu\bar{h}_{\mu\nu} \to \partial^\mu\bar{h}_{\mu\nu} + \partial^\mu\partial_\mu\xi_\nu\,. \tag{24}$$

Choosing ξ_ν to solve $\Box\xi_\nu = -\partial^\mu\bar{h}_{\mu\nu}$ (always possible: this is a sourced wave equation), we can impose the **Lorenz (harmonic / de Donder) gauge**

$$\partial^\mu\bar{h}_{\mu\nu} = 0\,. \tag{25}$$

In this gauge the second term, $\partial^\rho \partial_{(\mu}\bar{h}_{\nu)\rho} = \frac{1}{2}\left(\partial_\mu \partial^\rho \bar{h}_{\nu\rho} + \partial_\nu \partial^\rho \bar{h}_{\mu\rho}\right)$, and the third term, $\propto \partial^\rho \partial^\sigma \bar{h}_{\rho\sigma}$, both vanish, leaving

$$\Box\, \bar{h}_{\mu\nu} \equiv \partial^\rho \partial_\rho \bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu}\,. \tag{26}$$

In vacuum ($T_{\mu\nu} = 0$) this is the wave equation $(-\partial_t^2 + \nabla^2)\,\bar{h}_{\mu\nu} = 0$, whose solutions are waves propagating at the speed of light, e.g. plane waves $\bar{h}_{\mu\nu} = C_{\mu\nu} e^{ik_\rho x^\rho}$ with $k^\rho k_\rho = 0$ and $k^\mu C_{\mu\nu} = 0$. These ripples of the metric are gravitational waves. (Residual gauge transformations with $\Box\xi_\mu = 0$ can further be used to reach transverse-traceless gauge, $h_{0\mu} = 0$, $h = 0$, $\partial^i h_{ij} = 0$, leaving two physical polarisations h_+ and $h_\times$, as in part b.)

b) The full metric is $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, i.e.

$$ds^2 = -dt^2 + (1 + h_+(t-z))\,dx^2 + (1 - h_+(t-z))\,dy^2 + dz^2\,. \tag{27}$$

This perturbation is in transverse-traceless gauge ($h_{0\mu} = 0$, $h^\mu{}_\mu = h_+ - h_+ = 0$, and $\partial^i h_{ij} = 0$ since h_{ij} depends only on $t - z$ and has no z-components).

The test particles stay at fixed coordinates. A particle initially at rest has $u^\mu = dx^\mu/d\tau = (1,0,0,0)$. Its geodesic equation gives, to first order in h,

$$\frac{du^\mu}{d\tau} = -\Gamma^\mu{}_{\nu\rho} u^\nu u^\rho = -\Gamma^\mu{}_{00}\,, \qquad \Gamma^\mu{}_{00} = \frac{1}{2}\eta^{\mu\lambda}\left(\partial_0 h_{\lambda 0} + \partial_0 h_{0\lambda} - \partial_\lambda h_{00}\right) = 0\,, \tag{28}$$

since $h_{0\mu} = 0$. Hence $du^\mu/d\tau = 0$ and the particle remains at rest, $u^\mu = (1,0,0,0)$, for all time: freely falling particles keep fixed spatial coordinates in TT gauge (and $\tau = t$). The coordinate separations are therefore constant, $\Delta x = L_x$ and $\Delta y = L_y$; the physical effect shows up in the proper distance.

Particle at $(L_x, 0, 0)$. The proper distance at fixed time t along the straight line $y = z = 0$ from $x = 0$ to $x = L_x$ (on which $dt = dy = dz = 0$) is

$$\Delta s = \int_0^{L_x} \sqrt{g_{xx}}\, dx = \int_0^{L_x} \sqrt{1 + h_+(t-0)}\, dx = L_x \sqrt{1 + h_+(t)} \approx L_x \left(1 + \frac{1}{2} h_+(t)\right), \tag{29}$$

where we used that along the path $z = 0$ so $h_+(t-z) = h_+(t)$ is independent of x, and $\sqrt{1+\epsilon} \approx 1 + \epsilon/2$ to first order in the small perturbation. (To this order the straight coordinate line is the relevant spatial geodesic; corrections are $\mathcal{O}(h^2)$.)

Particle at $(0, L_y, 0)$. Similarly, along $x = z = 0$ from $y = 0$ to $y = L_y$,

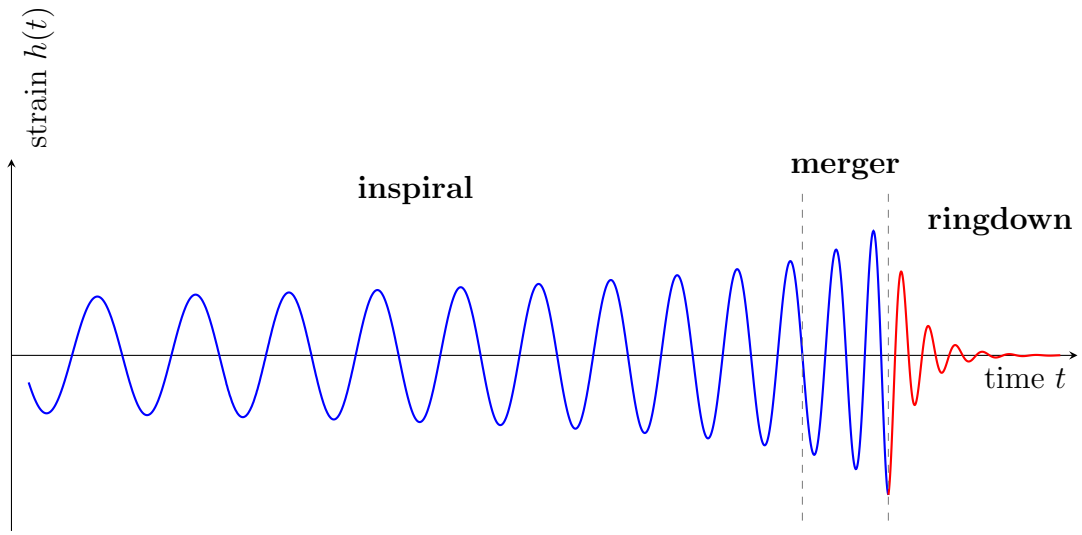
$$\Delta s = \int_0^{L_y} \sqrt{g_{yy}}\, dy = \int_0^{L_y} \sqrt{1 - h_+(t)}\, dy = L_y \sqrt{1 - h_+(t)} \approx L_y \left(1 - \frac{1}{2} h_+(t)\right). \tag{30}$$

So as the wave passes, the separation in the x direction is stretched while that in the y direction is squeezed (and vice versa half a period later), with fractional change $\pm\frac{1}{2}h_+(t)$.

c) **Qualitative evolution.** The gravitational waves carry energy and angular momentum away from the binary. Losing energy, the orbit shrinks: for a Keplerian circular orbit of separation r and total mass M, $E = -Gm_1m_2/(2r)$ and $\omega_{\rm orb}^2 = GM/r^3$, so as E decreases r decreases and the orbital frequency increases. The dominant (quadrupole) gravitational wave has frequency $f_{\rm GW} = 2f_{\rm orb}$ (the mass quadrupole returns to itself after half an orbit), and amplitude $h \sim \frac{(G\mathcal{M})^{5/3}(\pi f_{\rm GW})^{2/3}}{D}$ (with $\mathcal{M} = (m_1m_2)^{3/5}/M^{1/5}$ the chirp mass and D the distance), which also grows. Since the emitted power rises steeply as the orbit tightens ($P \propto \omega^{10/3}$), the process runs away: a "chirp" of increasing frequency and amplitude, with $\dot{f}_{\rm GW} = \frac{96}{5}\pi^{8/3}(G\mathcal{M})^{5/3}f_{\rm GW}^{11/3}$, so measuring f and $\dot{f}$ determines the chirp mass. The signal has three phases:

- **Inspiral:** the long, slow, quasi-circular phase in which the bodies spiral together adiabatically through many orbits. The waveform is a nearly sinusoidal oscillation whose frequency and amplitude both slowly increase (the chirp); well described by the quadrupole formula and post-Newtonian theory.
- **Merger:** once the bodies reach separations of order a few times GM (around the innermost stable circular orbit) they plunge and coalesce into a single, highly distorted black hole. This is the strong-field, highly nonlinear regime (modelled with numerical relativity); the signal reaches its maximum amplitude and frequency here.
- **Ringdown:** the distorted remnant settles down to a stationary Kerr black hole by radiating away its deformations as quasinormal modes. The signal is an exponentially damped sinusoid, $h \propto e^{-t/\tau}\cos(2\pi f_{\rm QNM}t)$, whose frequency and damping time are fixed purely by the final mass and spin (no-hair theorem). After this the signal is gone.

Sketch of the observed strain $h(t)$:



The frequency and amplitude increase through the inspiral (chirp), peak at merger, and then decay exponentially at fixed frequency in the ringdown. For a stellar-mass binary such as GW150914 (about $30M_\odot + 30M_\odot$) the frequency sweeps from $\sim 35\,\mathrm{Hz}$ to $\sim 250\,\mathrm{Hz}$ in a fraction of a second, within LIGO's sensitive band.

Detection by an interferometer. LIGO is a Michelson interferometer with two perpendicular arms of length $L = 4\,\mathrm{km}$, which we can take along the x and y axes, exactly as in part b): the beam splitter at the origin and the end mirrors at $(L, 0, 0)$ and $(0, L, 0)$ are suspended so that, in the horizontal plane and in the detector's frequency band, they behave as freely falling test masses. Laser light is split, travels down both arms (bouncing many times in Fabry–Pérot cavities to increase the effective path), and recombines at the beam splitter; the interferometer is set so that with no wave the light interferes destructively at the output photodetector. When a +-polarised wave travels along z (perpendicular to the detector plane), part b) shows that the proper length of the x arm becomes $L(1 + \frac{1}{2}h_+)$ while that of the y arm becomes $L(1 - \frac{1}{2}h_+)$, alternating as the wave oscillates. The arm-length difference

$$\Delta L(t) = L_x - L_y = h_+(t)\, L \tag{31}$$

changes the light travel times in the two arms and hence the relative phase of the recombined beams, $\Delta\phi \simeq 2{\cdot}2\pi\, \Delta L/\lambda$ (per round trip), so light leaks to the photodetector with an intensity that tracks $h(t)$. The detector thus directly reads out the strain $h(t)$, i.e. the waveform sketched above (more generally $h = F_+ h_+ + F_\times h_\times$ with antenna pattern factors depending on the source direction and polarisation). Typical strains are $h \sim 10^{-21}$, so $\Delta L \sim 10^{-18}\,\mathrm{m}$, which is why extreme isolation from seismic, thermal and quantum noise is needed, and why two (or more) widely separated detectors are used to confirm coincident signals.

Connecting signal and physics: the inspiral's frequency evolution $\dot{f}(f)$ gives the chirp mass; the frequency at merger ($\sim$ the ISCO frequency $\propto 1/M$) gives the total mass; the ringdown frequency and damping time give the mass and spin of the final black hole; and the overall amplitude (given the masses) gives the distance D to the source.